

Luca Nunziante

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WORK EXPERIENCE

Robot Learning Researcher

Jun. 2024 – Present

Araya Inc.

Tokyo, Japan

- Designed and executed a real-robot evaluation of six Vision-Language-Action (VLA) models, collecting tens of hours of teleoperation data and training and deploying every variant, resulting in ~5,000 rollouts.
- Open sourcing all the policy rollouts across six models and spanning more than 20 robot hours. Top contributor at oopsie-data.com.
- Developed statistical analyses of task performance and trajectory quality beyond standard success-rate metrics.
- Built imitation-learning pipelines for robot manipulation, fine-tuning and deploying image- and point-cloud-based policies on physical robots across multiple projects.
- Open-sourced a VLA batched inference and kv-cache implementation that improved throughput by up to 40× for the DSRL algorithm.
- Implemented operational-space and Control Barrier Function-based controllers and teleoperation interfaces, enabling safe data collection and repeatable deployment of learned policies on physical robots.
- Developed custom MuJoCo simulation environments and robot-control interfaces used throughout a user study with 20+ participants.

Visiting Research Student

Dec. 2023 – Apr. 2024

Tohoku University

Sendai, Japan

- Developed compliant control methods for robotic assembly tasks and a Model Predictive Control scheme for collision avoidance.
- Built a dual-arm solar-panel assembly pipeline as an MSc thesis project, leading to a peer-reviewed publication at SII 2025; project developed in ROS.

PUBLICATIONS

[†] Equal contribution.

When One Step Is Enough: An Empirical Study of Euler Integration in Flow-Matching Policies

Under Review

S. Sujit[†], L. Nunziante[†], et al.

An Augmented Reality Brain-Robot Interface for Generalist Robot Arm Manipulation

RO-MAN 2026

S. Zhang, R.F.J. Dossa, L. Nunziante, et al.

[Website]

Levels of Shared Autonomy in Brain-Robot Interfaces: Enabling Multi-Robot Multi-Human Collaboration for Activities of Daily Living

Frontiers 2026

H. Douglas[†], M. Di Vincenzo[†], R.F.J. Dossa[†], L. Nunziante[†], et al.

[Website]

Improving Low-Cost Teleoperation: Augmenting GELLO with Force

SII 2025

S. Sujit[†], L. Nunziante[†], et al.

Assembling Solar Panels by Dual Robot Arms Towards Full Autonomous Lunar Base Construction

SII 2025

L. Nunziante, et al.

[Video]

Energy-Based Residual for Collision Detection Using a Velocity Observer

I-RIM 2024

G. Gravina[†], L. Nunziante[†], et al.

[Video]

RESEARCH PROJECTS

Policy gradient RL on VLAs

Jul. 2026

- Running policy gradient RL on large VLAs on OGBench dataset.

VLA Latent Space Investigation

Jan. 2026

- Reproduced a feature-based VLA failure detector and evaluated latent-space interventions for modifying policy behavior in simulation. Measured their impact on task success and found no consistent improvement in task success rate.

Advantage-Conditioned VLA Learning

Dec. 2025

- Implemented a RECAP-style training pipeline using a pretrained VLM to generate zero-shot value estimates from rollout observations, converted the estimates into advantages, and fine-tuned a VLA conditioned on those advantages.

TECHNICAL SKILLS

Programming: Python, C++, MATLAB

Libraries / Frameworks: PyTorch, MuJoCo, ROS, Eigen

Tools: Git, Isaac Gym, Gazebo

EDUCATION

Sapienza University of Rome

Rome, Italy

MSc in Artificial Intelligence and Robotics — 110/110 cum laude

2021 – 2024

- Thesis: “Autonomous Assembly of Solar Panels by Dual Robot Arms in the Remote Space.”
- Supervisors: Prof. A. De Luca and Prof. K. Yoshida.

University of Campania Luigi Vanvitelli

Caserta, Italy

BSc in Electronic and Computer Engineering — 110/110 cum laude

2018 – 2021

- Thesis: “Object Detection with Neural Networks for Robotics Applications: Training Methods Compared.”
- Supervisor: Prof. C. Natale.

LANGUAGES

Italian: Mother tongue

English: Professional working proficiency

Japanese: Conversational